

HDOC - 32

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Question

Write a C program to perform the following menu-driven operations on a given 2D array:

1. Find the sum of all elements in a matrix.
2. Find the sum of each row of a matrix and store it in an array.
3. Find the transpose of a matrix.
4. Exit

Program

```
#include <stdio.h>

int main() {
    int a[10][10], rowSum[10];
    int rows, cols, i, j, choice, sum;

    printf("Enter the number of rows and columns: ");
    scanf("%d %d", &rows, &cols);

    printf("Enter the matrix elements:\n");
    for (i = 0; i < rows; i++)
        for (j = 0; j < cols; j++)
            scanf("%d", &a[i][j]);

    do {
        printf("\n--- MENU ---\n");
        printf("1. Sum of all elements\n");
        printf("2. Sum of each row\n");
        printf("3. Transpose of matrix\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                sum = 0;
                for (i = 0; i < rows; i++)
                    for (j = 0; j < cols; j++)
                        sum += a[i][j];
                printf("Sum of all elements = %d\n", sum);
                break;

            case 2:
                printf("Row sums:\n");
                for (i = 0; i < rows; i++) {
                    rowSum[i] = 0;
                    for (j = 0; j < cols; j++)
                        rowSum[i] += a[i][j];
                    printf("Sum of row %d = %d\n", i + 1, rowSum[i]);
                }
                printf("The row-sum array contains: [");
                for (i = 0; i < rows; i++) {
                    printf("%d", rowSum[i]);
                    if (i < rows - 1) printf(", ");
                }
                printf("]\n");
                break;

            case 3:
                printf("Transpose of the matrix:\n");
                for (j = 0; j < cols; j++) {
                    for (i = 0; i < rows; i++)
                        printf("%d\t", a[i][j]);
                }
                break;

            default:
                break;
        }
    } while (choice != 4);
}
```

```

        printf("\n");
    }
    break;

case 4:
    printf("Exiting program...\n");
    break;

default:
    printf("Invalid choice\n");
}
} while (choice != 4);

return 0;
}

```

Sample Output

Enter the number of rows and columns: 2 3

Enter the matrix elements:

1 2 3

4 5 6

Enter your choice: 1

Sum of all elements = 21

Enter your choice: 2

Row sums:

Sum of row 1 = 6

Sum of row 2 = 15

The row-sum array contains: [6, 15]

Enter your choice: 3

Transpose of the matrix:

1 4

2 5

3 6

Enter your choice: 4

Exiting program...

Result: The program successfully performs all required menu-driven operations on the 2D array.